

Module 7

Assignment 7B: Basic Routing

Course: CS 85 - PHP Programming

Instructor: Vicky Seno

Project Overview

This project is a four-page personal Laravel website. It demonstrates closure-based and controller-based routes, data passed into Blade views, a reusable master layout, a dynamic route parameter, named navigation, active-link styling, and 404 error handling.

The standalone application lives in assignments/module7b and can also be opened from the main CS85 Module 7 roadmap. This app-inside-an-app structure keeps the submission complete while connecting it to the larger coursework portfolio.

Route List

URL	Route Name	Type	Purpose
/	home	Closure	Show the welcome page and pass my name and title to the view.
/about	about	Closure	Show information about me and pass student data to the view.
/hobbies	hobbies.index	Controller	List all personalized hobbies through HobbyController@index.
/hobbies/{id}	hobbies.show	Controller	Show one hobby by numeric ID or return a 404 response.

Customization

Personal information added: Siarhei Hancharou, Santa Monica College, and Web Development (A.S.). The About page also introduces my coursework platform, Lens Lounge microservices project, photography portfolio, and GitHub profile. It uses the privacy-safe value “Prefer not to disclose” for age until a numeric value is intentionally provided.

Photography: I selected photography because it combines technical camera skills, creativity, communication, and my SERGIOARTG portfolio work.

Web Development: I selected web development because I enjoy turning ideas into accessible interfaces and learning how frontend and backend systems work together.

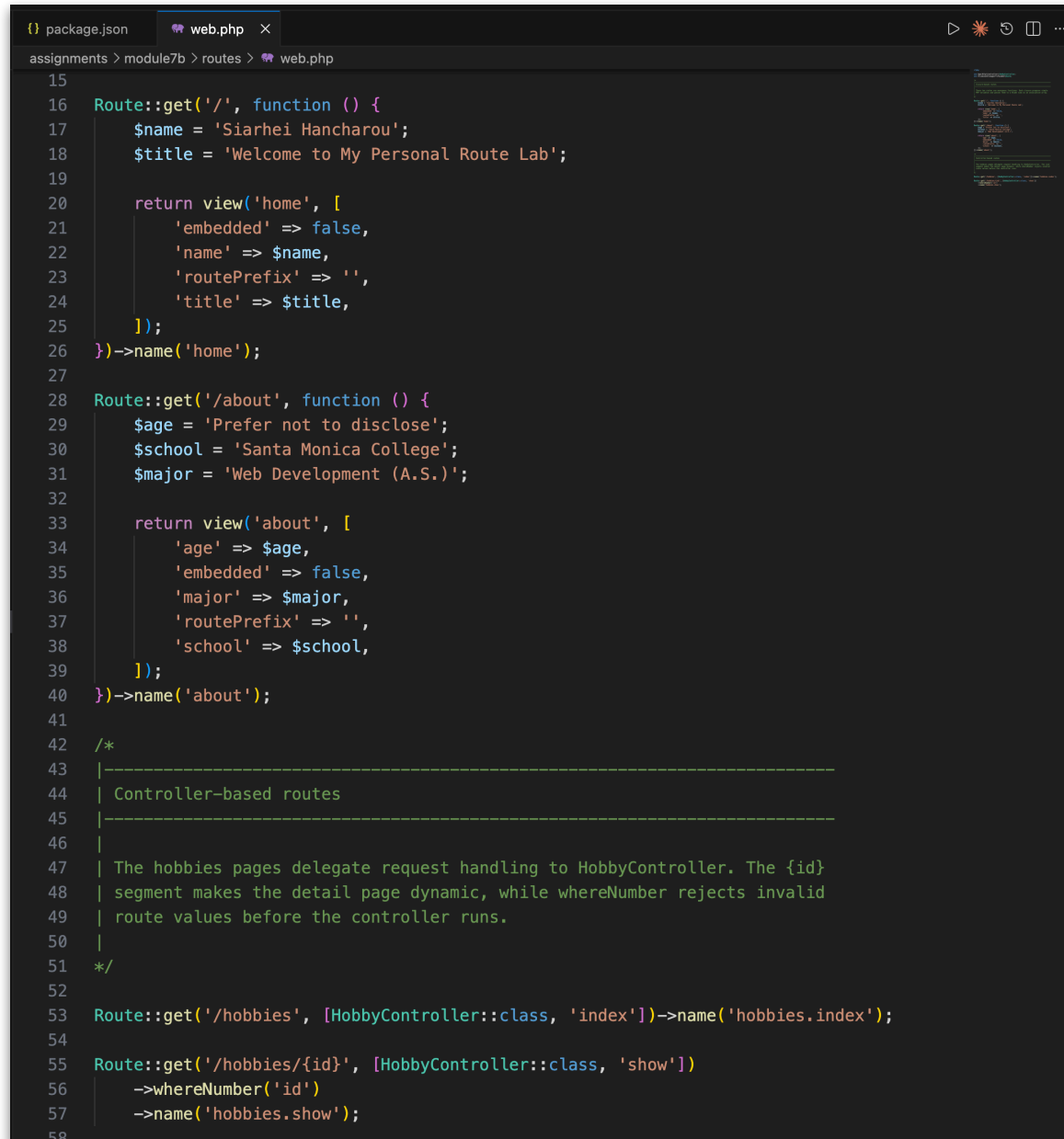
Technology Projects: I selected technology projects because hands-on experiments help me practice debugging, version control, documentation, and product presentation.

GitHub Repository: <https://github.com/sergehall/cs85-php-programming/tree/main/assignments/module7a>

Screenshots

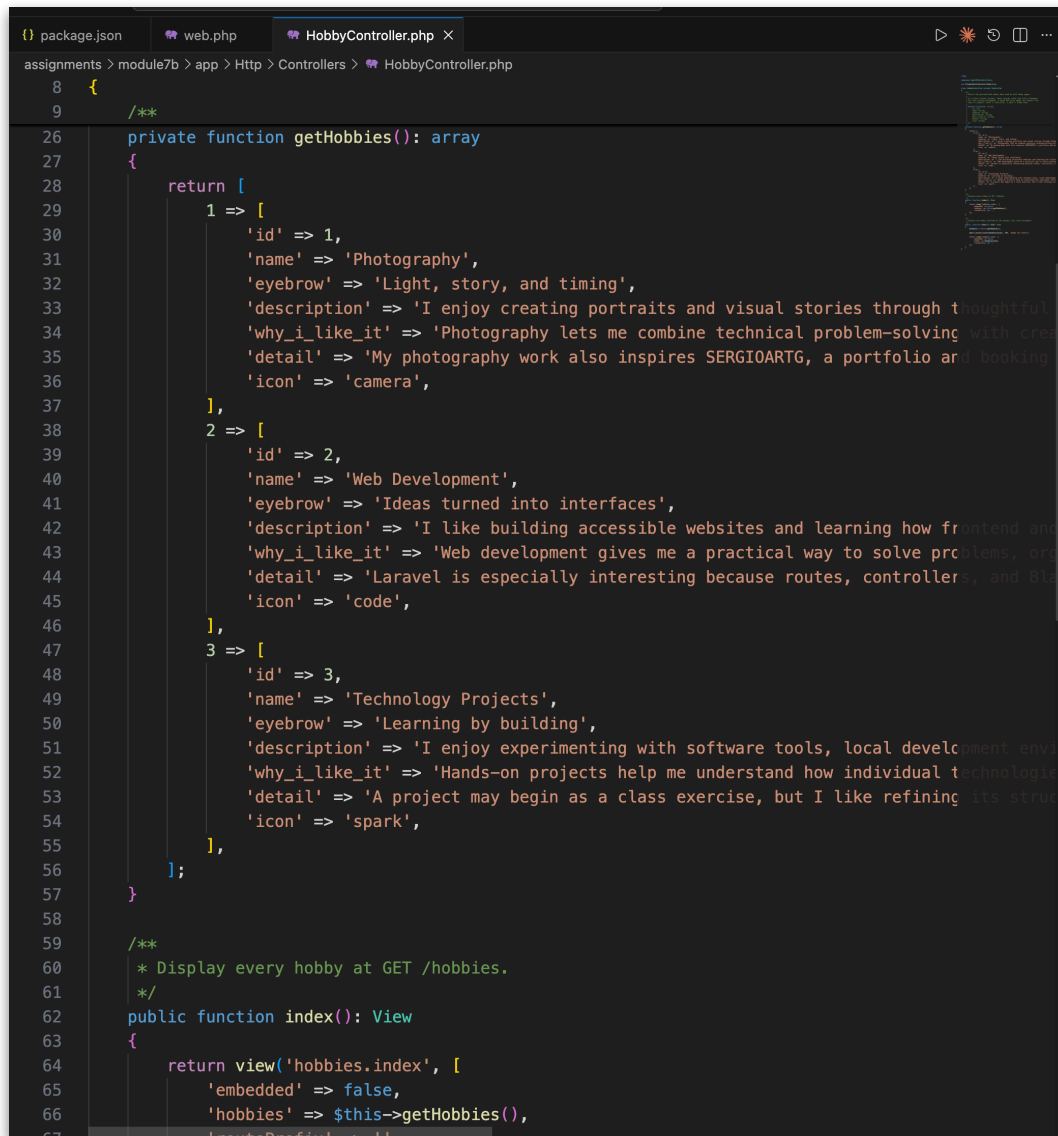
1. Routes File (routes/web.php)

The application defines four named routes. The Home and About pages use closure-based routes, while the Hobbies pages use a controller. Named routes are used throughout the application to simplify navigation and improve maintainability.



```
15
16 Route::get('/', function () {
17     $name = 'Siarhei Hancharou';
18     $title = 'Welcome to My Personal Route Lab';
19
20     return view('home', [
21         'embedded' => false,
22         'name' => $name,
23         'routePrefix' => '',
24         'title' => $title,
25     ]);
26 }->name('home');
27
28 Route::get('/about', function () {
29     $age = 'Prefer not to disclose';
30     $school = 'Santa Monica College';
31     $major = 'Web Development (A.S.)';
32
33     return view('about', [
34         'age' => $age,
35         'embedded' => false,
36         'major' => $major,
37         'routePrefix' => '',
38         'school' => $school,
39     ]);
40 }->name('about');
41
42 /*
43 |-----
44 | Controller-based routes
45 |-----
46 |
47 | The hobbies pages delegate request handling to HobbyController. The {id}
48 | segment makes the detail page dynamic, while whereNumber rejects invalid
49 | route values before the controller runs.
50 |
51 */
52
53 Route::get('/hobbies', [HobbyController::class, 'index'])->name('hobbies.index');
54
55 Route::get('/hobbies/{id}', [HobbyController::class, 'show'])
56     ->whereNumber('id')
57     ->name('hobbies.show');
```

2. HobbyController (HobbyController.php)



```
8  {
9  /**
26 private function getHobbies(): array
27 {
28     return [
29         1 => [
30             'id' => 1,
31             'name' => 'Photography',
32             'eyebrow' => 'Light, story, and timing',
33             'description' => 'I enjoy creating portraits and visual stories through thoughtful
34             'why_i_like_it' => 'Photography lets me combine technical problem-solving with crea
35             'detail' => 'My photography work also inspires SERGIOARTG, a portfolio and booking
36             'icon' => 'camera',
37         ],
38         2 => [
39             'id' => 2,
40             'name' => 'Web Development',
41             'eyebrow' => 'Ideas turned into interfaces',
42             'description' => 'I like building accessible websites and learning how frontend and
43             'why_i_like_it' => 'Web development gives me a practical way to solve problems, org
44             'detail' => 'Laravel is especially interesting because routes, controllers, and Bla
45             'icon' => 'code',
46         ],
47         3 => [
48             'id' => 3,
49             'name' => 'Technology Projects',
50             'eyebrow' => 'Learning by building',
51             'description' => 'I enjoy experimenting with software tools, local development envi
52             'why_i_like_it' => 'Hands-on projects help me understand how individual technologie
53             'detail' => 'A project may begin as a class exercise, but I like refining its struc
54             'icon' => 'spark',
55         ],
56     ];
57 }
58
59 /**
60  * Display every hobby at GET /hobbies.
61  */
62 public function index(): View
63 {
64     return view('hobbies.index', [
65         'embedded' => false,
66         'hobbies' => $this->getHobbies(),
67         'routePrefix' => ''
```

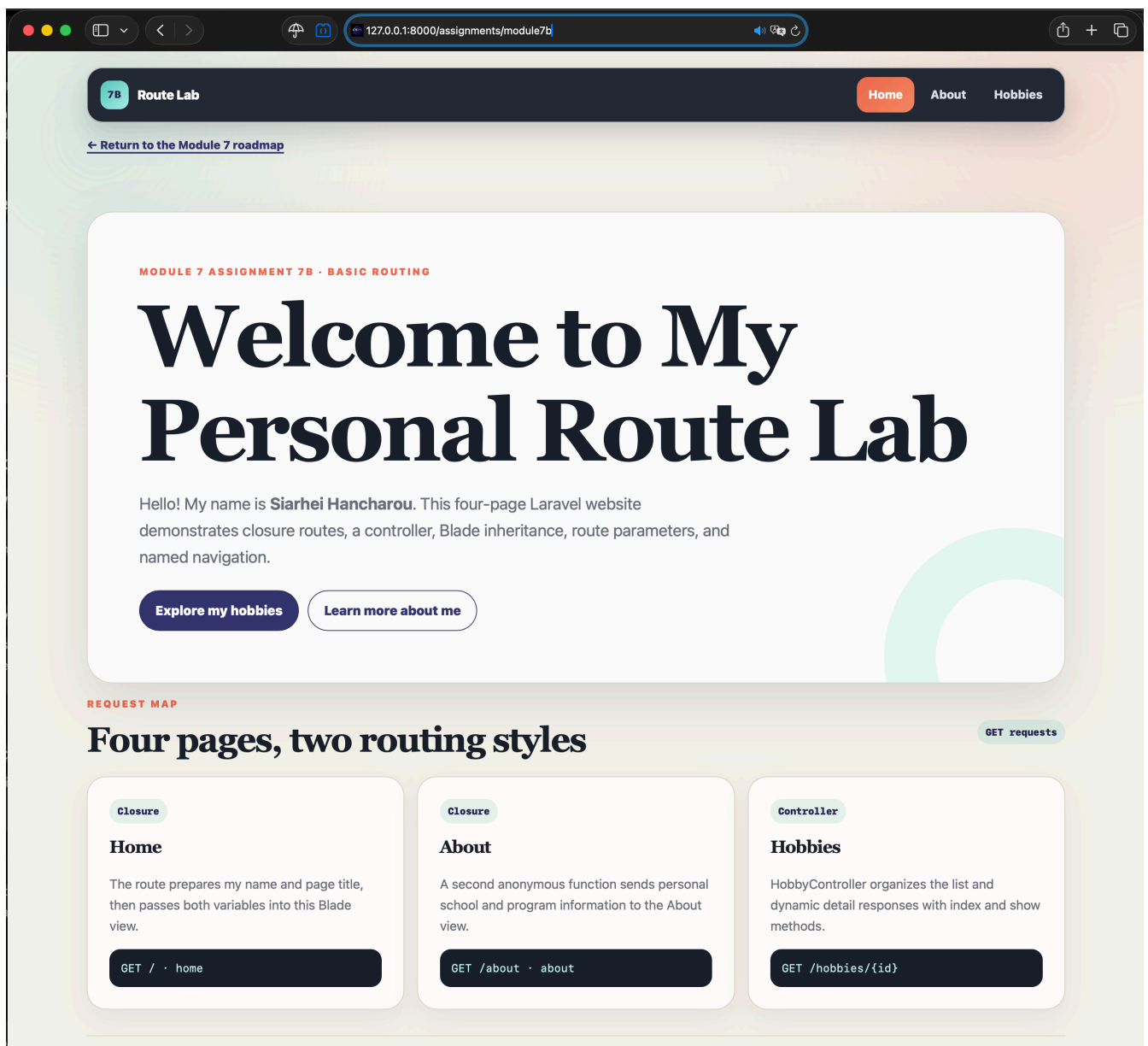
The controller contains two public methods:

- index() displays the list of hobbies.

- show(\$id) displays information about a selected hobby.

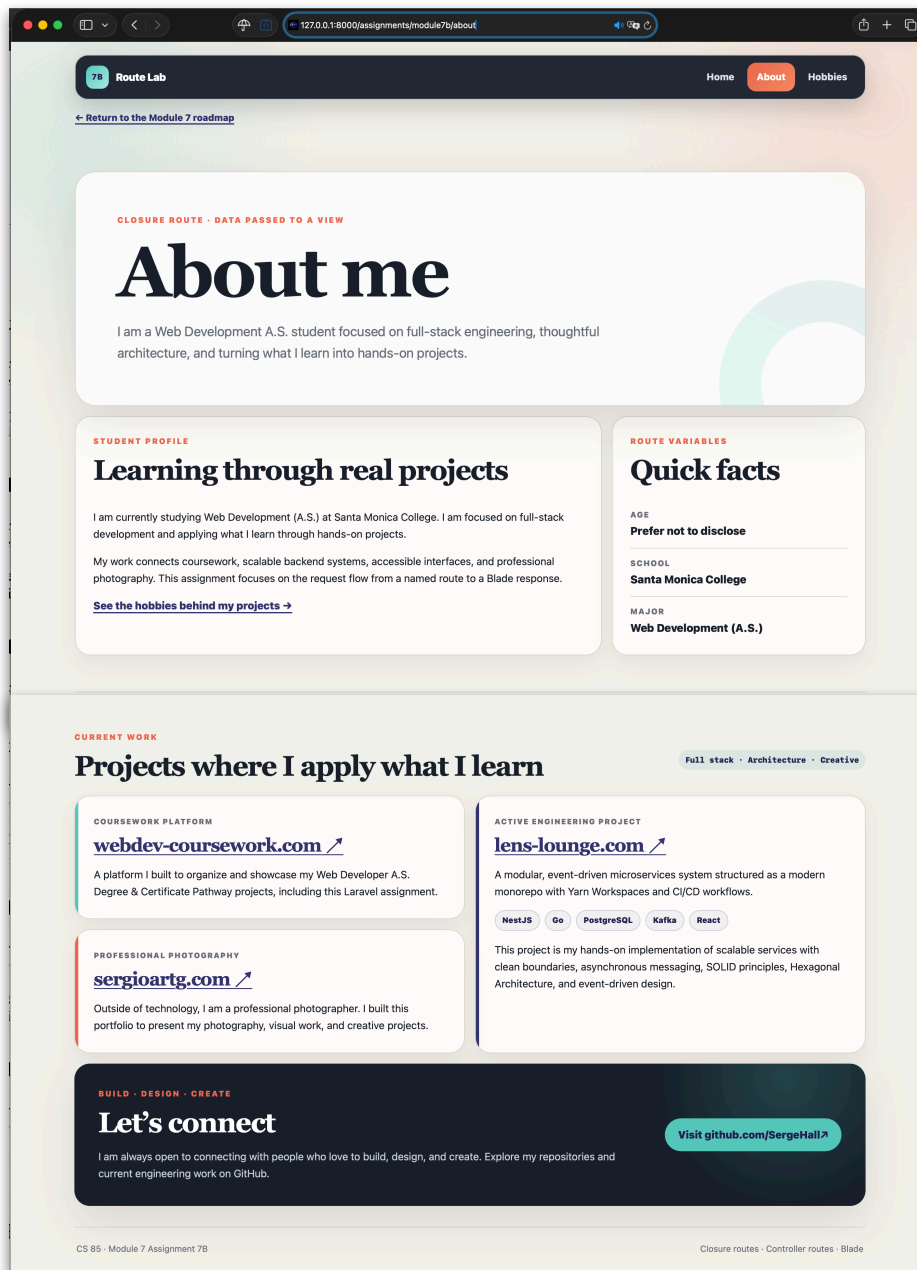
The hobby data is stored in a private helper method and passed to the Blade views.

3. Home Page



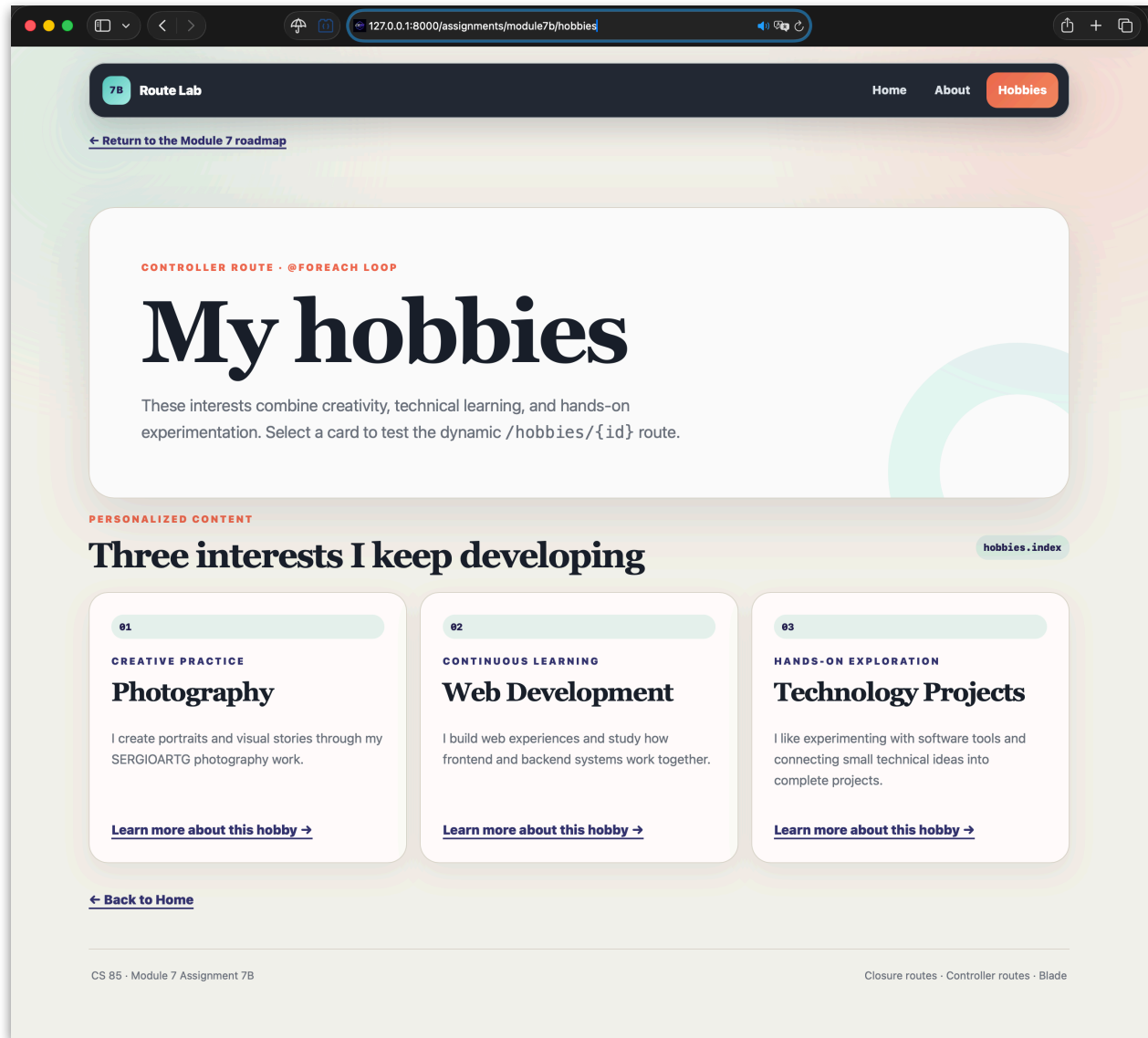
The Home page welcomes visitors and displays my name. It also provides navigation links to the About and Hobbies pages using Laravel named routes.

4. About Page



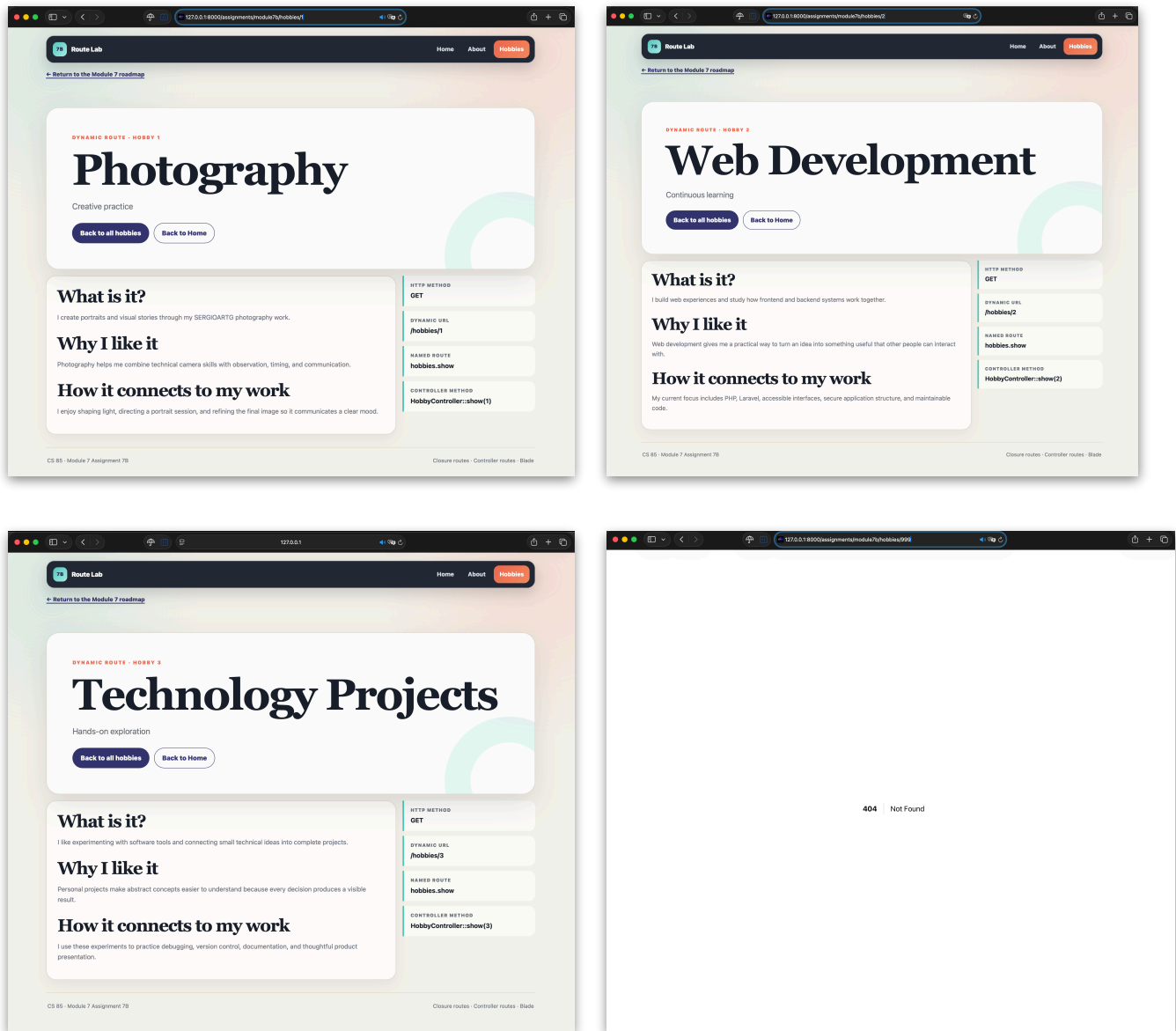
The About page contains personal information including my age, school, and major. It also includes a link to the hobbies section.

5. Hobbies Page



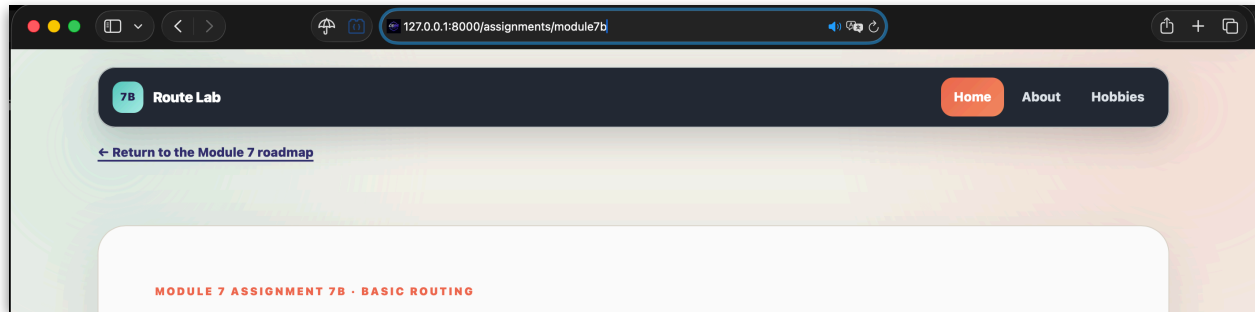
The Hobbies page displays all hobbies using a Blade `@foreach` loop. Each hobby includes a short description and a link to its detail page.

6. Application Hobbies Pages



The screenshots of the Home, About, Hobbies, Hobby Detail, and Laravel 404 Error pages demonstrate that all required pages are functioning correctly. The application successfully displays each page, and an invalid hobby ID correctly returns the Laravel 404 error page, confirming that the routing and error handling work as expected.

7. Navigation Menu



The navigation menu uses Laravel named routes. The active page is highlighted, making navigation clear and user-friendly.

Customization

Personal Information Added

I customized the website by adding my own personal information:

- **Name:** Siarhei Hancharou
- **School:** Santa Monica College
- **Major:** Web Development (A.S.)

Instead of displaying a numeric age, I chose to use “Prefer not to disclose.” This still demonstrates how personal data can be created inside a Laravel route, passed to a Blade view, and displayed dynamically on the page.

I also added links to my coursework platform, software projects, photography portfolio, and GitHub profile. These details make the website more personal and connect the assignment to my real educational and professional interests.

My Hobbies

1. Photography

Photography is one of my favorite creative activities. I enjoy creating portraits and visual stories by working with lighting, composition, timing, and editing. It allows me to combine technical camera skills with creativity and communication.

My photography experience also connects to SERGIOARTG, the portfolio and booking platform I built to present my creative work online.

2. Web Development

Web development allows me to combine creativity with problem-solving. I enjoy building accessible websites and learning how frontend and backend technologies work together.

Laravel is especially interesting to me because routes, controllers, and Blade templates create a clear path from a browser request to the final page shown to the user.

3. Technology Projects

I enjoy experimenting with programming tools, development environments, and practical software projects. Working on hands-on projects helps me understand how individual technologies connect and gives me experience with debugging, testing, documentation, and version control.

A project may begin as a classroom assignment, but I enjoy improving its structure and presentation until it becomes a stronger portfolio project.

Reflection

What was the easiest part of this assignment?

The easiest part was creating and personalizing the Blade views after the routes were connected. Blade template inheritance made it simple to use one master layout for all pages while keeping the navigation, styling, and footer consistent.

It was also easy to display personal data with Blade's `{{ }}` syntax after passing variables from a route or controller to a view.

What was the most challenging part?

The most challenging part was understanding the complete request flow for the dynamic hobby route. The `{id}` parameter begins in the URL, is passed to the `show()` method in `HobbyController`, and determines which hobby is sent to the Blade view.

Testing an invalid URL such as `/hobbies/999` also helped me understand why the controller should verify that a record exists before attempting to display it.

What did you learn about Laravel routing?

This assignment helped me understand how Laravel connects an HTTP method and URL to a closure or controller method. Closure-based routes work well for simple pages, while controllers make related request logic easier to organize and maintain.

I also learned how to pass data to Blade views, use dynamic route parameters, create named routes, and build navigation with Laravel's `route()` helper. Named routes make the application easier to update because the Blade templates do not need to repeat hard-coded URLs.